INCH-POUND
MIL-PRF-3098/146D
7 January 2010
SUPERSEDING
MIL-PRF-3098/146C(CR)
19 December 2003

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR165/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.4 MHz to 20 MHz; fundamental; noncontrolled; antiresonance.

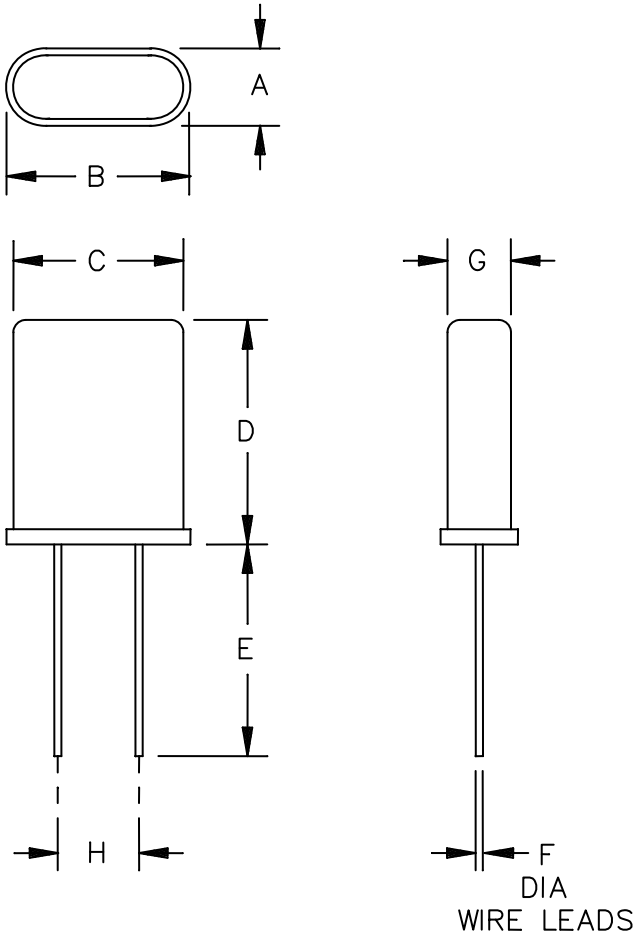

FIGURE 1. Crystal unit - CR165/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.183	---	4.65
B	---	.440	---	11.18
C	---	.403	---	10.24
D	---	.530	---	13.46
E	.500	---	12.70	---
F	.016	.019	0.41	0.48
G	---	.150	---	3.81
H	.184	.200	4.67	5.08

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Corrosion resistant metal plating of the enclosure after resistance welding is permitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR165/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.4 MHz to 20.0 MHz, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive. MHz	Maximum resistance Ohms
2.40 to 2.60	300
2.60+ to 2.90	250
2.90+ to 3.75	180
3.75+ to 4.75	120
4.75+ to 6.00	75
6.00+ to 7.00	50
7.00+ to 10.00	30
10.00+ to 20.00	25

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW maximum.

Antiresonance: load capacitance: 30 pF ±0.5 pF.

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Capacitance, shunt:

2.4 MHz to 4.0 MHz: 1.7 pF minimum to 7.0 pF maximum.
4.0 MHz to 6.0 MHz: 2.7 pF minimum to 7.0 pF maximum.
6.0 MHz to 20.0 MHz: 3.7 pF minimum to 7.0 pF maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Temperature cycling:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): CR165/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5955-2009-080)

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